

Biology

Chapter 3.1: Transport in Molecules

Welcome to AnithUncommon's Sample Resources!

We are assuming you are reading this as a student. So What will **YOU** Expect From This Sample?

In Biology Chapter 3.1, you will explore how molecules can move from one place to another, specifically within a living organism

This sample will be divided into **three** parts: Introduction to Samples (this part), Lesson Objectives (What is Expected From Students), and most crucially, Syllabus Contents.

Note that the 'Syllabus Content' does NOT stand as a complete syllabus itself; rather, it is a tight compilation of notes gathered by Kenneth V. to create a comprehensive sample for students.

If you want a complete version of these notes and think that you need help in Biology, join our nonprofit!

Gmail: anithuncommon

Instagram handle: @anithuncommon

This sample is made thanks to, Kenneth V.

— Made by students, for students.

Lesson Objectives:

At the end of the lesson, students should be *able* to...

- Name the ways molecules can be transported within a living organism
- Explain how each method of transportation works

Reflective Questions to Consider:

All cells need to be able to move crucial molecules in and around itself. If that's the case (and it is), the following questions are highlighted:

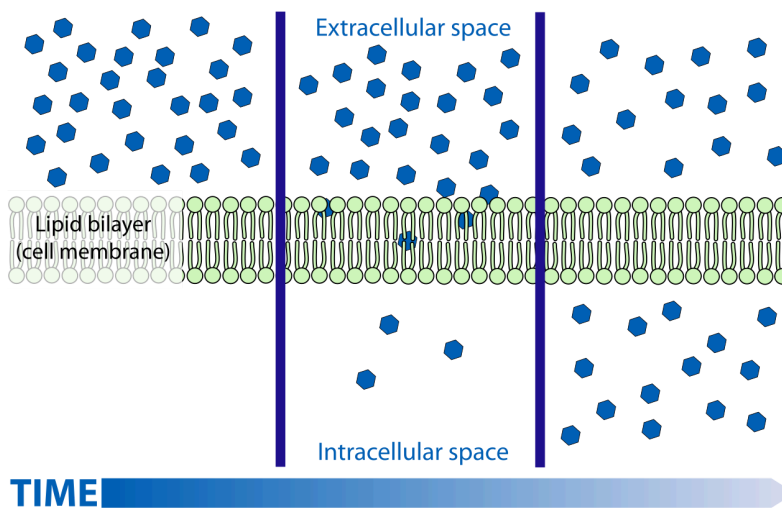
- How do molecules move around within a living organism's body?
- What differences do the methods used for transportation have?

Chapter 3: Transport in Organisms	
Tip! • It's good to realize that how molecules/particles move is almost universal. This means that if there's an increase of kinetic energy, they're most likely going to move faster around Activities in Class: • Lesson Breakdown • Kahoot-type quiz	
1.1 . Transport of Molecules	Mentor's Notes
Simple Diffusion	These methods of

A. Description

→ Net **movement** of **particles** from a region of **higher** concentration to a region of **lower** concentration as a result of **random movement**

→ Diffusion is one of the most common methods where molecules can move around (ex.g. You can smell food when you walk by food stands)



B. Affecting factors

→ It is affected by its **concentration gradient**, the steeper it is = the faster it goes. In other other words, if site A is high in concentration, and site B is low in concentration; The lesser the concentration of site B, the faster the molecules move from site A to site B

→ As **temperature** increases, the molecules gain more **kinetic energy** and vibrate/ **move faster** overall, allowing a faster rate of diffusion

→ The **nature of the particles** used within a diffusion matters, as the **heavier/ larger/ polar** particles are **harder** to move than lighter/ smaller/ non-polar particles

→ The **surface area** where the molecules diffuse

transportation form the basic understanding of how molecules move into and out of a cell or perhaps anywhere in some cases. It is important to realize how this is useful for a cell's function (especially active transport, facilitated diffusion, and bulk transport).

Each method of transportation has its own unique characteristics that play into the cell's needs. For example, on neurons, a protein pump (carrier protein) uses active transportation to stimulate a nerve impulse. This happens due to the pump moving 3 sodium ions out, and 2 potassium ions in (This process is known as co-transportation),

through also affect the rate of diffusion, as the more surface area = the more spaces where molecules can diffuse through

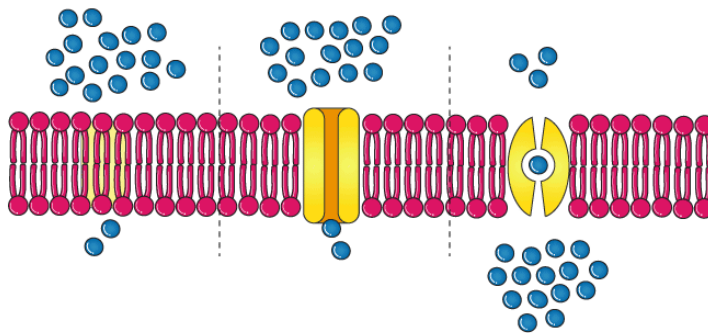
creating a potential difference.

Facilitated Diffusion

A. Description

→ **Diffusion** of a substance through a **transport protein** (Channel/Carrier) in a cell membrane

→ Facilitated diffusion is most commonly found on the cellular level, specifically on the **cell surface membranes** where molecules/ particles need to be pushed out or pulled in



B. Affecting Factors

→ Factors affecting facilitated diffusion are similar to simple diffusion

→ A primary difference is how many **transport proteins** are **available** to move particles. In other words, if all transport proteins are used up (clogged), they won't be able to take in more particles hence reducing the rate of diffusion

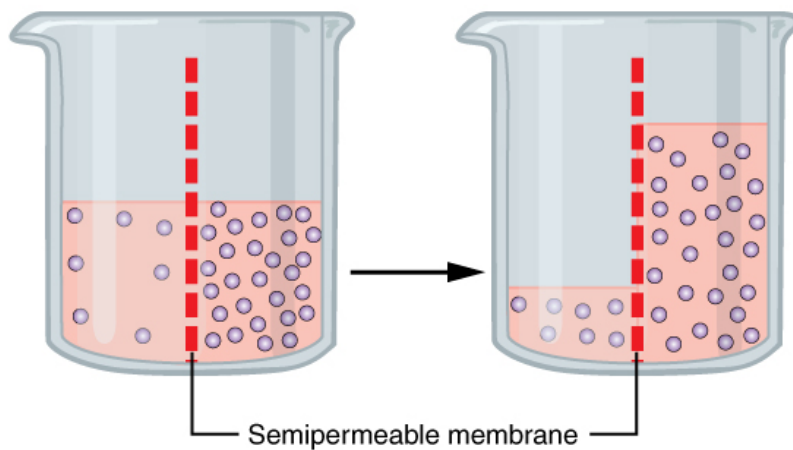
Water Potential & Osmosis

A. Description

→ Water potential is the **measure** of **tendency** for **water** to move from **one place** to **another** (kPa,

symbol of Ψ) [Pure Water is 0 kPa]

→ **Osmosis** is the net **diffusion** of **water** molecules from a region of **less negative** water potential (**less solute**) to a region of **more negative** water potential (**more solute**), through a **partially permeable membrane**



B. Affecting Factors

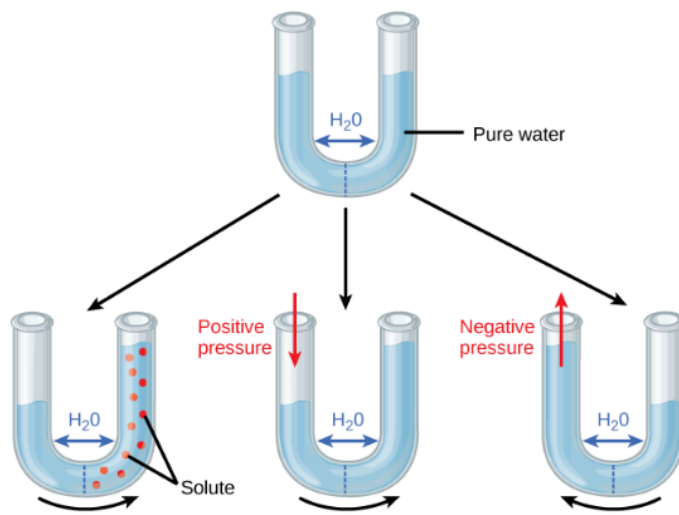
→ With an increase of **positive pressure** (pressure being given towards the direction of osmosis), the water potential **increases** due to energy being given that forces water through

→ With an increase of **negative pressure** (pressure being given opposing the direction of osmosis), the water potential **decreases** due to energy being 'taken' that forces water out

→ As **solute** concentration **increases**, water potential **decreases** due to less water molecules being able to move as they bind to solutes

→ The bigger the difference between water potentials, the faster the rate of osmosis (**Steeper concentration gradient**)

→ Similar to diffusion, **temperature** also has an effect

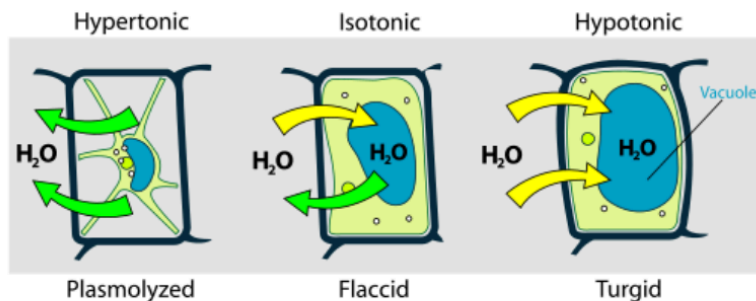


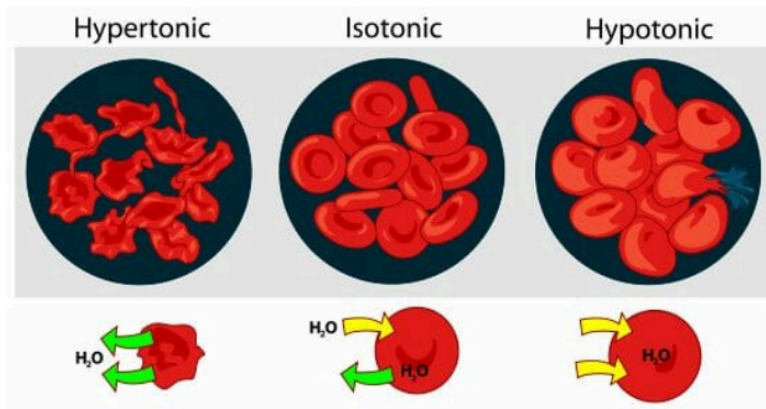
C. Effects on cells

→ Osmosis is an important phenomenon to consider when looking at cells

→ If a cell is surrounded within a solution with a water potential more negative than inside the cell (**Hypertonic Solution**), water will **leave** the **cell** to equalize water potentials

→ On the contrary, if a cell is surrounded within a solution with a water potential less negative than inside the cell (**Hypotonic Solution**), water will enter the cell to equalize water potentials. This process might increase **turgor pressure** and **swell** the cell (**Plants**) or completely **burst** the cell due to overpressure (**Animals**)

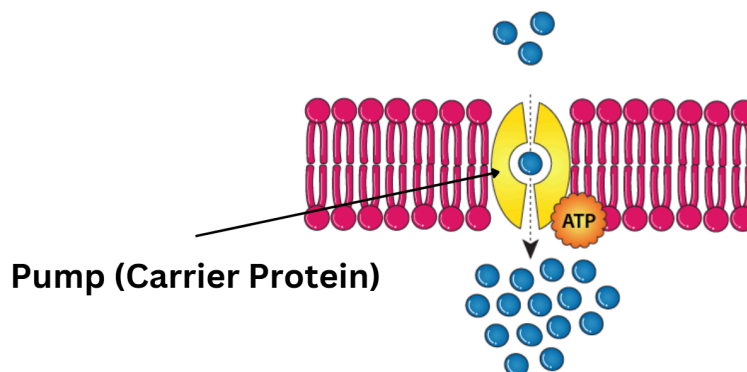




Active Transport

A. Description

- Net movement of **particles** across a **cell membrane** going **against** their **concentration gradient**
- It usually uses **energy (ATP)** to be able to push the particles against the concentration gradient



B. Affecting Factors

- Since active transport is dependant on pumps, if there's no **ATP** to power the pumps or if no **pumps** are available, the rate decreases
- **Temperature** and the **steepness** of the concentration gradient also affect the rate

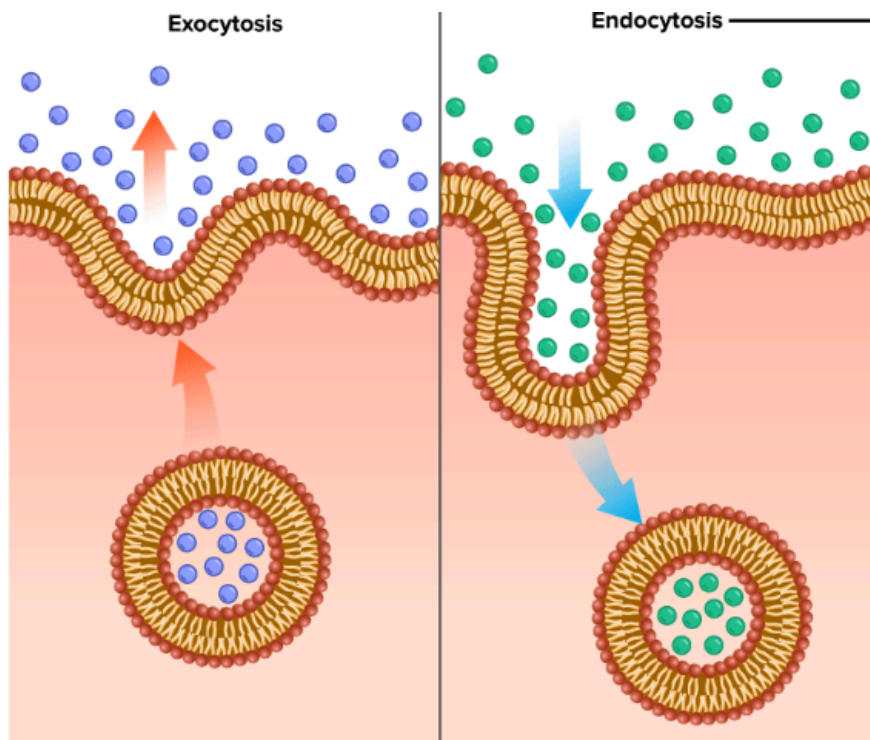
Bulk/ Vesicular Transport

A. Description

→ Net **movement** of **particles** across a **cell membrane** using **membrane-bound vesicles** that uses **energy**

→ It is split into two types: **Exocytosis** (Out) and **Endocytosis** (In)

→ Endocytosis is split further into two types: **Phagocytosis** (Uptake of solids) and **Pinocytosis** (Uptake of liquids)



Comparison of Different Transport Methods

	Type	Direction	Substance Moved	Requirement
Simple Diffusion	Passive	High to Low	Small, non-polar particles	N/A

Facilitated Diffusion	Passive	High to low	Large/Polar	Transport protein	
Osmosis	Passive	High to Low (Water Potential)	Water only	Semi-permeable membrane	
Active Transport	Active	Low to High	Ions, Nutrients	Pump & ATP	
Vesicle Transport	Active	N/A	Large particles, proteins	Vesicles	

Mentors' Sources:

- AS & A Level Biology Blue Book
- Save My Exams
- Khan Academy